Shift Cipher Attacks Documentation

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In this project I updated the previous insecure chat application by implementing an old symmetric key encryption algorithm called **Caesar Cipher**. This encryption algorithm is one type of shift algorithm where letters of the alphabet is shifted by some number(in this case for Ceasar Cipher shift = 3).

So, In this project Bob and Alice exchange encrypted data and decrypt it after receiving the message which make no sense for anyone who is sniffing the packet. But there is a catch.

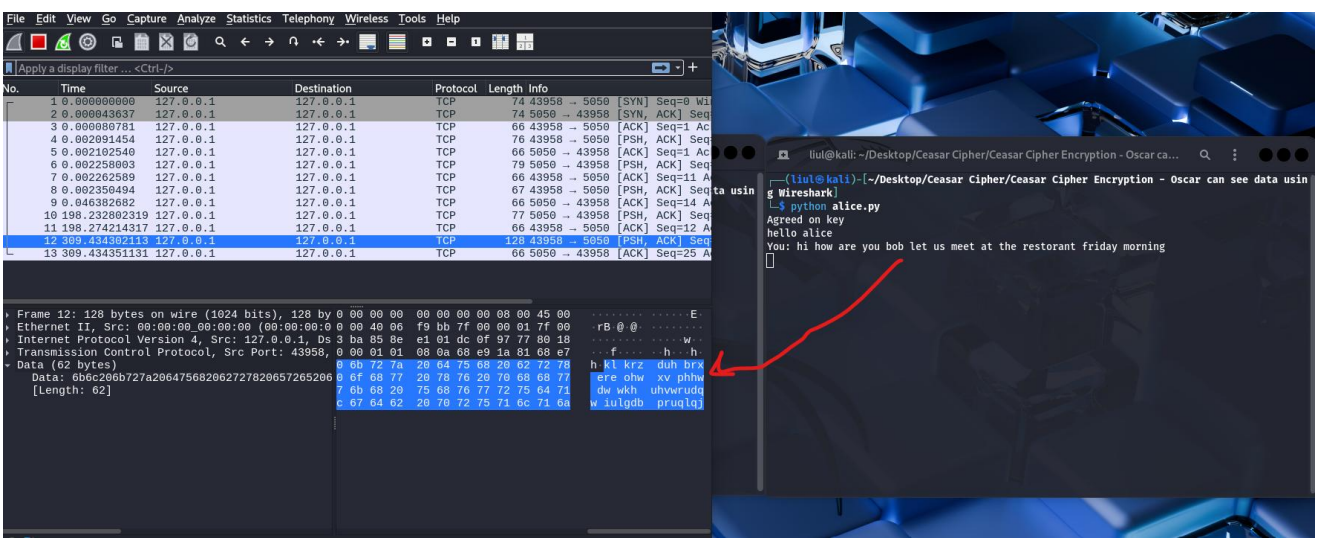
If the threat actor(Oscar) gets the first message which is key exchange, even though he gets scrambled message using Wireshark, he will use the key to decrypt it and read the message just like the intended individuals.

This is the key weakness of symmetric key encryption algorithms.

What Oscar sees (Oscar is the bad guy)

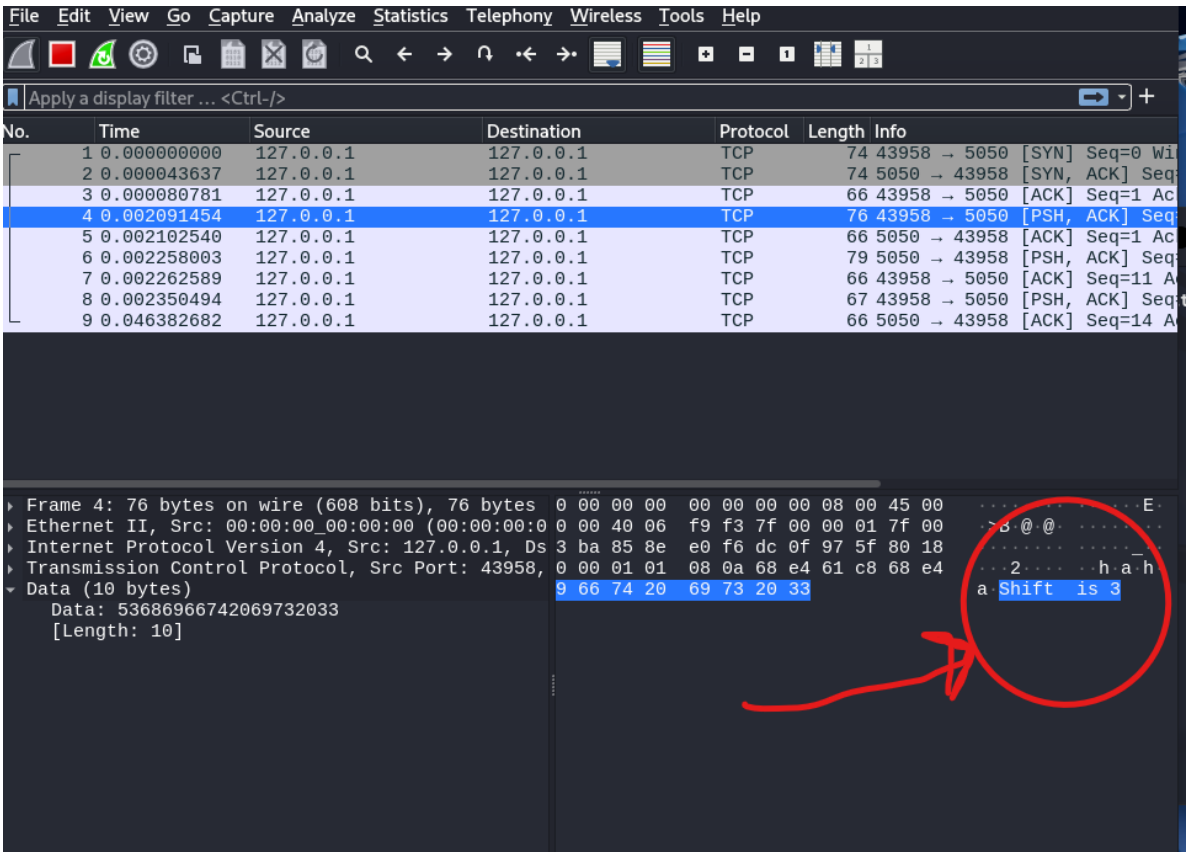
If Oscar doesn't have the key, all he sees is just something that doesn't make any sense.

Bob and Alice are exchanging meaningful messages whereas Oscar is confused and is asking what the heck is going?



If bob found the Key, Game Over!!!

If Oscar find the key which is the first message that is exchanged between Bob and Alice, right after the TCP handshake. That key exchange packet is unencrypted, it is in plain text. It is used for later encryption. This is the weakness of **Symmetric Encryption Algorithm**.



Normal Conversation Between Alice and Bob(in windows)

Even though Bob and Alice are communicating perfectly, Oscar is capturing the packets and decrypting the messages with out them knowing.

```
PS D:\education\self\python\Encrypted vs Unencrypted Chatting Application\Encrypted> python bob.py
listening on port 5050
connected to ('127.0.0.1', 64060)

you: hello there
hello i am liul
you: where should we meet
meet me at the restorant
you: at what time
seven
you: okay
bye
you: bye
PS D:\education\self\python\Encrypted vs Unencrypted Chatting Application\Encrypted>
```

What Did I learn?

- Python Sockets in detail
- Oldest Symmetric Encryption Algorithm(Ceasar Cipher)
- Analyzing packets using packet analyzer(Wireshark)
  Used loopback address
- Understanding Symmetric Encryption has Weakness
- Understanding that even if Oscar doesn't have the shift key, lets say they exchanged the key in person, Oscar can use Brute Force Attack as the key space is small = 26